

ACTIVE PASS FILTER

Aim

To design and set up the active First order filters using Op-amp.

Apparatus Required

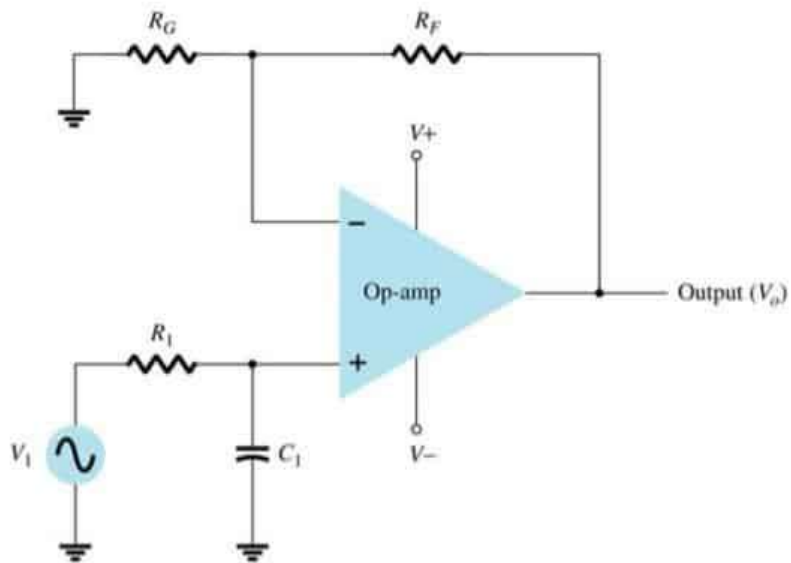
Op-amp, resistors, capacitors, breadboard, CRO, function generator and power supplies.

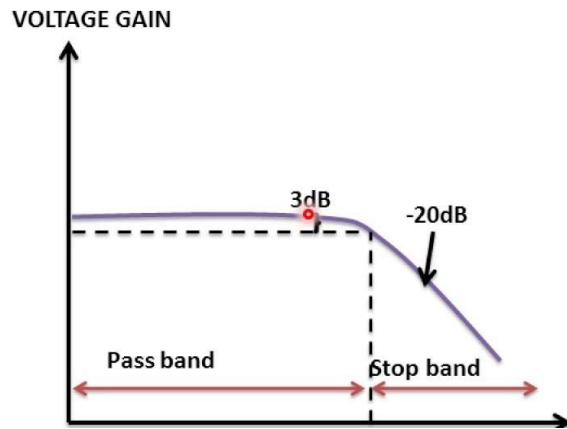
Design:

$$F_c = \frac{1}{2\pi RC}$$

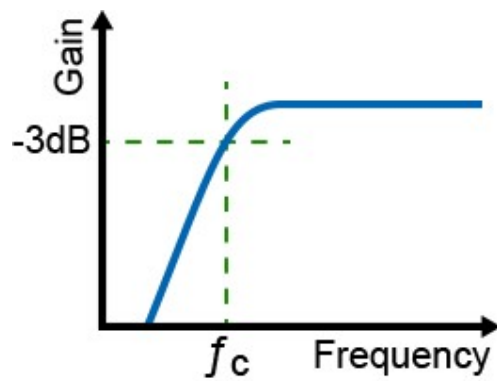
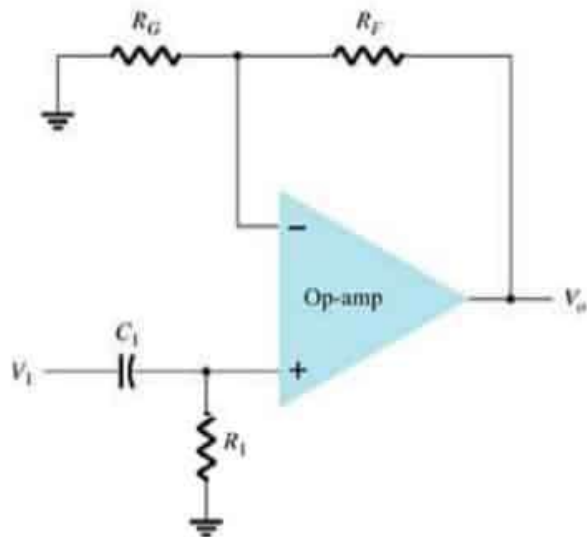
Circuit Diagram

First Order LPF





First Order HPF

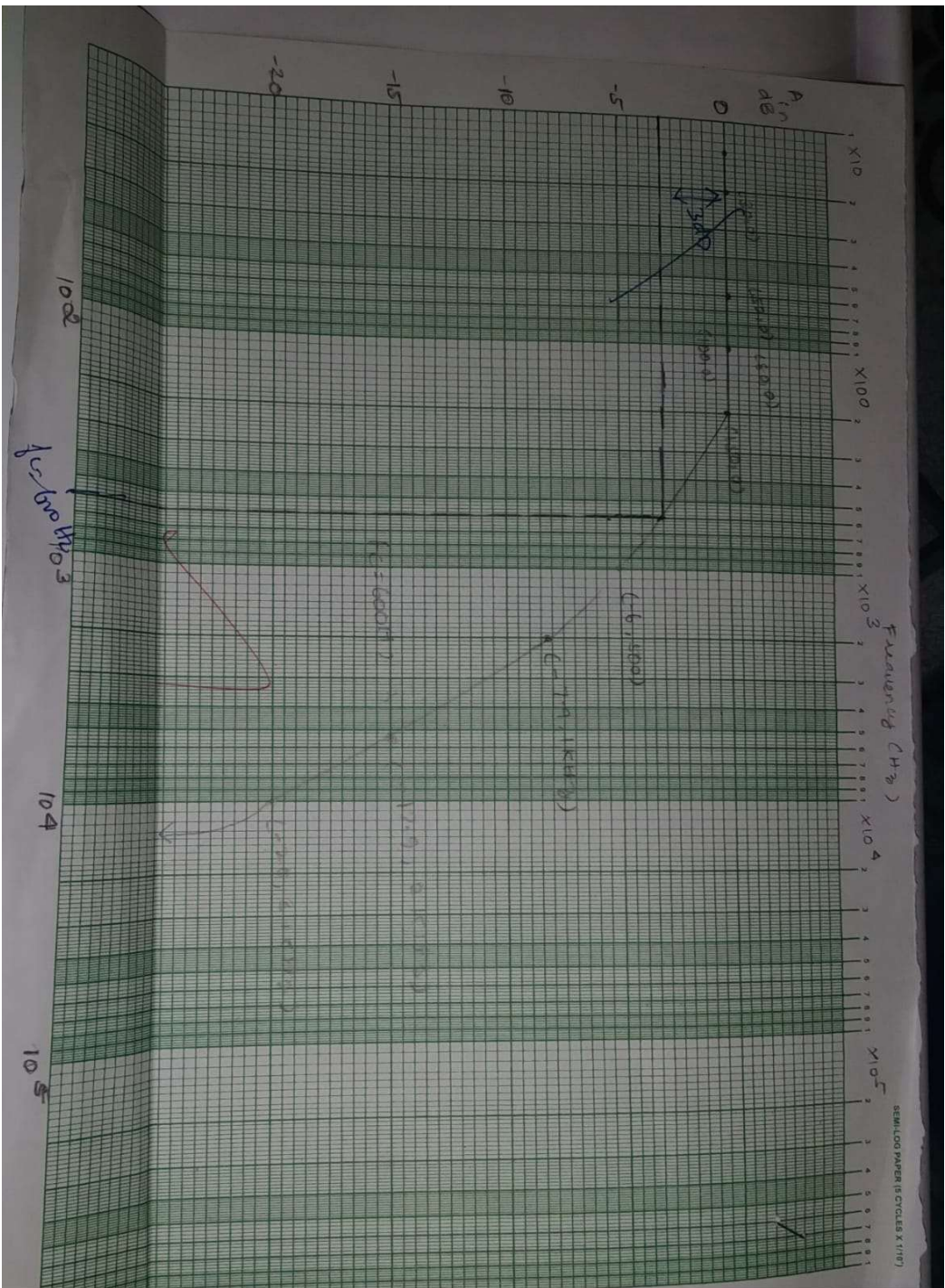


Procedure

1. Assemble the first-order active filter circuit on the breadboard.
2. Connect IC 741 to a **$\pm 15\text{ V DC}$ supply**.
3. Connect the function generator to the input of the filter.
4. Apply a sinusoidal input, for example, **1 V Vm**
5. Initially set the input frequency well below the calculated cut-off frequency.
6. Observe V_{in} and V_{out} on the CRO/DSO.
7. Measure the output voltage at different input frequencies.
8. Calculate the voltage gain IN Decibel.
9. Increase the frequency gradually through and beyond the expected cut-off frequency.
10. Record the output voltage at each frequency.
11. Identify the frequency at which the gain becomes approximately **-3 dB from the passband gain**.
12. Plot the frequency-response curve of **gain (dB) versus frequency**.
13. Compare the experimentally obtained cut-off frequency with the theoretical value.

Observation Table

S. No.	Frequency (Hz)	V_{in} (V)	$V_{out}(V)$	Gain	Gain (dB)
1	30	1	1	1	0
2	50	1	1	1	0
3	80	1	1	1	0
4	100	1	1	1	0
5	150	1	1	1	0
6	500	1	500mv	500	-6.02
7	1K	1	400mv	400	-7.9
8	2K	1	200mv	200	-13.9
9	3K	1	100mv	100	-20



Result:

Thus Designed the First order filters using op amp and obtained the results as Cutoff frequency of low pass filter is $f_c = 600$ Hz